

Assignment 1 cloud data engineering
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Question 1 — SELECT & WHERE

SQL

```
SELECT first_name, last_name, city, phone  
FROM sales.customers  
WHERE state = 'CA'  
AND phone IS NOT NULL;
```

Question 2 — ORDER BY (Multiple Columns)

SQL

```
SELECT product_id, product_name, model_year, list_price  
FROM production.products  
ORDER BY model_year DESC, list_price ASC;
```

Question 3 — TOP N & TOP PERCENT

◆ Part a

SQL

```
SELECT TOP 5 product_name, list_price  
FROM production.products  
ORDER BY list_price DESC;
```

◆ Part b

SQL

```
SELECT TOP 5 PERCENT *  
FROM production.products  
ORDER BY list_price ASC;
```

```
-- Row count depends on total rows  
-- Example: If total rows = 100 → returns 5 rows  
-- Example: If total rows = 200 → returns 10 rows
```

Question 4 — OFFSET & FETCH (Pagination)

◆ Page 1 (Rows 1–10)

SQL

```
SELECT product_id, product_name, list_price  
FROM production.products  
ORDER BY list_price DESC  
OFFSET 0 ROWS FETCH NEXT 10 ROWS ONLY;
```

- ◆ Page 2 (Rows 11–20)

SQL

```
SELECT product_id, product_name, list_price
FROM production.products
ORDER BY list_price DESC
OFFSET 10 ROWS FETCH NEXT 10 ROWS ONLY;
```

- ◆ Page 3 (Rows 21–30)

SQL

```
SELECT product_id, product_name, list_price
FROM production.products
ORDER BY list_price DESC
OFFSET 20 ROWS FETCH NEXT 10 ROWS ONLY;
```

Question 5 — DISTINCT

- ◆ Part a

SQL

```
SELECT DISTINCT state
FROM sales.customers
ORDER BY state ASC;
```

- ◆ Part b

SQL

```
SELECT DISTINCT state, city
FROM sales.customers
ORDER BY state ASC, city ASC;
```

- ◆ Part c

SQL

```
SELECT COUNT(DISTINCT model_year) AS total_model_years
FROM production.products;
```

Question 6 — Logical Operators (AND / OR)

SQL

```
SELECT product_id, product_name, brand_id, category_id, list_price
FROM production.products
WHERE list_price BETWEEN 500 AND 1500
      AND (model_year = 2019 OR model_year = 2020)
ORDER BY list_price ASC;
```